

DANBURY, CT, USA

WMO: 725086

Lat: 41.371N

Lon: 73.483W

Elev: 457

StdP: 14.45

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 54734

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.8	9.2	-9.7	3.3	8.5	-5.6	4.1	13.4	20.4	31.3	18.6	30.5	4.6	290	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	88.3	72.9	85.6	71.7	83.1	70.4	75.8	83.9	74.4	81.6	73.0	79.7	8.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	125.7	79.7	72.2	121.2	78.4	70.8	115.6	77.0	39.5	84.0	38.1	81.5	36.8	79.9	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	16.1	13.8	-3.7	93.4	5.5	2.9	-7.7	95.4	-10.9	97.1	-14.0	98.8	-18.0	100.9
			-3.9	78.9	5.8	1.5	-8.0	79.9	-11.4	80.8	-14.6	81.6	-18.8	82.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	27.1	29.6	37.2	48.6	58.4	66.8	72.0	70.6	64.0	52.4	42.0	33.0		
	DBStd	17.47	10.39	9.18	8.93	8.14	7.43	6.30	5.06	5.13	6.64	8.01	8.09	9.07		
	HDD50	2685	712	573	408	119	10	0	0	0	0	67	263	533		
	HDD65	6008	1176	992	862	496	232	53	6	11	99	399	689	993		
	CDD50	2774	1	1	11	77	271	505	683	637	419	141	23	5		
	CDD65	624	0	0	0	4	28	109	224	184	68	7	0	0		
	CDH74	5119	0	0	1	82	331	927	1878	1384	474	42	0	0		
	CDH80	1360	0	0	0	21	80	239	567	350	98	5	0	0		
(14) Wind	WSAvg	5.8	6.2	6.4	7.1	6.9	5.9	5.4	4.9	4.5	4.7	5.5	5.7	5.9		
(15) Precipitation	PrecAvg	49.3	3.5	3.1	4.3	4.1	3.9	4.3	4.3	4.6	4.7	4.2	4.2	4.2		
	PrecMax	65.0	7.3	8.0	10.7	8.8	6.6	9.1	8.0	10.7	12.6	15.0	6.9	6.2		
	PrecMin	33.0	1.7	1.5	0.9	1.6	0.9	1.1	1.5	1.9	0.9	0.3	1.5	1.3		
	PrecStd	7.5	1.4	1.4	2.1	1.8	1.4	2.2	1.9	2.2	2.9	3.0	1.3	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	60.1	69.7	83.2	87.1	90.1	92.0	91.0	87.7	78.9	68.1	63.5		
		MCWB	54.8	53.3	55.6	63.3	69.0	73.6	75.3	75.3	72.4	66.3	59.3	55.9		
	2%	DB	53.3	53.6	62.6	74.8	82.1	85.9	88.7	86.7	82.5	73.4	63.7	56.9		
		MCWB	49.4	47.5	51.0	58.4	66.8	71.7	73.7	73.0	69.7	65.2	57.7	52.8		
	5%	DB	46.3	48.0	56.6	69.2	77.6	82.3	85.8	83.7	79.3	70.1	60.7	52.4		
		MCWB	42.3	41.6	47.1	55.7	64.4	69.9	72.1	71.3	68.4	63.1	54.6	48.7		
	10%	DB	42.1	43.2	51.9	63.8	72.9	79.5	82.9	81.4	75.8	66.2	56.5	47.4		
		MCWB	37.4	37.5	44.6	52.8	61.8	67.6	70.9	69.8	66.9	59.9	51.1	43.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.8	56.0	58.8	65.5	72.1	76.0	78.2	77.9	75.3	70.9	63.1	58.9		
		MCDB	57.7	59.0	65.0	78.6	83.1	85.5	88.2	87.1	82.7	74.7	66.3	61.8		
	2%	WB	49.6	48.0	53.1	62.0	69.2	74.1	76.1	75.5	73.0	67.6	59.4	53.6		
		MCDB	52.5	51.8	58.7	72.4	78.0	82.7	84.9	82.3	78.6	72.1	62.6	56.3		
	5%	WB	42.9	41.9	48.7	58.4	66.9	72.3	74.4	74.1	71.3	64.5	56.0	48.9		
		MCDB	46.4	46.5	55.2	66.0	74.5	80.0	81.8	80.6	76.4	68.7	59.4	52.4		
	10%	WB	37.8	37.9	44.8	54.5	64.3	70.2	73.0	72.4	69.3	61.1	52.0	43.6		
		MCDB	41.3	42.6	50.7	61.0	70.6	76.5	79.8	78.1	73.9	65.1	55.4	47.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.5	17.8	18.2	20.8	20.5	19.4	19.3	18.8	19.9	19.3	18.4	15.6		
		MCDBR	20.7	23.8	25.9	29.7	26.7	22.9	22.0	21.0	22.5	22.9	22.8	20.9		
		MCWBR	17.9	18.3	16.8	15.8	13.7	11.5	9.7	9.6	11.9	14.3	17.1	18.6		
	5% WB	MCDBR	20.6	22.0	22.6	25.4	22.5	20.1	18.1	16.8	17.0	18.5	19.3	20.2		
		MCWBR	18.7	18.5	17.1	15.2	12.8	11.1	9.4	8.7	10.4	13.6	17.1	19.1		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.315	0.340	0.367	0.411	0.462	0.476	0.453	0.381	0.347	0.318	0.302		
	taud		2.442	2.422	2.421	2.387	2.286	2.178	2.152	2.205	2.381	2.494	2.530	2.520		
	Ebn at Noon		270	284	288	287	275	260	255	257	271	269	264	262		
	Edh at Noon		25	29	33	36	42	46	47	43	34	27	23	21		
(44) All-Sky Solar Radiation	RadAvg		582	859	1177	1504	1677	1833	1887	1665	1366	923	646	482		
	RadStd		44	78	91	159	163	167	103	106	122	93	55	48		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (70 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+110

Nomenclature: See separate page